Scale, Shape and Synthesis: a Reproducible Single-GPU Study of Zero-Shot Depth, Feed-Forward Reconstruction, Gaussian Splatting and Next-Frame Synthesis on nuScenes-mini

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Code: https://github.com/pavanyadava007/depth3d-lab

Abstract

We run four kinds of pretrained image-to-3D models on the same frames of one public driving data set, nuScenes-mini (10 scenes, 404 keyframes, six cameras and a LiDAR), on a single NVIDIA L4 GPU. The aim is to see how far geometry estimated from images can be trusted and at which step it breaks down. Three zero-shot monocular depth models (Depth Pro, Depth Anything V2 and MoGe-2) get the shape of a driving scene about right but miss its metric scale to very different degrees. Scaling each image by its median brings Depth Pro's AbsRel against LiDAR from 0.257 down to 0.178, Depth Anything V2's from 0.582 to 0.194 and MoGe-2's from 0.193 to 0.151. Depth Pro gets worse when it is given the true focal length, which we did not expect. MoGe-2 gets better. VGGT, a feed-forward multi-view model, has no metric scale, but with one scale per sequence it matches Depth Pro on the same LiDAR pixels (paired AbsRel difference -0.010 [-0.026, 0.007] on five-frame windows) and its relative rotations are accurate (median 0.14 degrees). The scale-aligned monocular depth also helps 3D Gaussian Splatting from sparse views. LiDAR initialisation gains 2.89 dB of held-out PSNR over random initialisation, and depth supervision adds another 2.22 dB. Synthesis is harder. Single-image Gaussians from SHARP reproduce the input view at 41.62 dB but drop to 20.42 dB at the true pose of the next keyframe, and a small flow-matching model for next-frame prediction is perceptually further from the real next frame than a plain copy of the current one (LPIPS 0.099 versus 0.040 at 0.10 s spacing). The study is small on purpose, with one data split, one machine and a few seeds. All tables and numbers are written by scripts that we release.

1 Introduction

Pretrained models can now estimate depth from one image, recover cameras and depth from a handful of images, and render new views from a single photograph. Most of them are evaluated on their own benchmarks, which makes it hard to see how they relate to each other. In this report we put several of them on the same frames of one driving data set and ask, at each stage, how much of the geometry they provide can be relied on. We use nuScenes-mini [2], a public split of ten 20-second scenes recorded with six cameras and a LiDAR, and we measure everything on one NVIDIA L4 GPU.

The work is organised in five tracks, and the later ones reuse results of the earlier ones. Track A compares monocular metric depth with LiDAR. Track B runs feed-forward multi-view reconstruction on the same pixels. Track C trains Gaussian Splatting with the scale-aligned depth maps from Track A as extra supervision. The last two tracks are about synthesis. In Track D we render single-image Gaussians at a true future pose, and in Track E we train a small conditional flow-matching model to predict the next frame.

Contributions.

- We split the error of zero-shot metric depth into a raw and a median-scaled part on all 2,424 nuScenes-mini images, broken down by camera, camera group and depth bin. Depth Pro and MoGe-2 are also run

with the true focal length, and our Depth Pro code is checked against the library's own inference path (Section 4.1).
- We compare VGGT and Depth Pro on the same LiDAR pixels of the same reference images, with three ways of aligning VGGT and bootstrap confidence intervals over sequences. Relative pose errors are measured against nuScenes ego poses (Section 4.2).
- We run a 3D Gaussian Splatting ablation on the most static scene, with random or LiDAR initialisation and with or without monocular depth supervision, at two step budgets and scored on held-out timestamps (Section 4.3).
- We measure how well single-image Gaussians from SHARP re-render the true next pose, and how a conditional flow-matching next-frame model compares with a copy-frame baseline (Sections 4.4 and 4.5).
- Every table and every number in the text is written by a script from the measured json files, so a rerun updates the paper. The scope is limited to nuScenes-mini, one NVIDIA L4, and 3 seeds for 3DGS and 3 for the next-frame models (Section 6).

2 Related work

Monocular metric depth. Depth Pro [1] predicts a canonical inverse depth and a field of view and combines the two into metric depth, so it needs no camera metadata. If the focal length is known it can be passed in and then replaces the model's estimate. Depth Anything V2 [13] is a relative-depth model that also comes in metric variants fine-tuned on metric labels. We use the outdoor metric Large model, which was fine-tuned on Virtual KITTI. MoGe-2 [10] adds a metric scale to the affine-invariant point maps of MoGe. Our evaluation protocol follows the KITTI convention [3, 4].

Feed-forward multi-view reconstruction. DUSt3R [11] regresses point maps for image pairs and aligns them globally. VGGT [9] replaces this with a single transformer that predicts cameras, depth and point maps for one to many views in one pass. Neither has a metric scale. We align VGGT in three ways: with one scale per sequence, with one scale per image, and with a similarity transform fitted to its camera centres [8].

Gaussian Splatting. 3D Gaussian Splatting [5] optimises anisotropic Gaussians that are normally initialised from an SfM point cloud. We use the gsplat rasteriser and its default densification strategy [14] with our own training loop. LiDAR points take the place of the SfM points, and a depth loss against the aligned monocular maps can be switched on.

Synthesis. SHARP [7] predicts metric 3D Gaussians from one image in a single forward pass. We render them at the true relative pose of later frames. Our next-frame model is trained with conditional flow matching [6]. Image quality is reported as PSNR, SSIM [12] and LPIPS [15].

3 Data and protocol

Data. nuScenes v1.0-mini [2] contains 10 scenes from Boston and Singapore, 404 keyframes at 2 Hz, six 1600×900 cameras and a 32-beam LiDAR, plus 12 Hz camera sweeps without annotations. nuScenes is released under CC BY-NC-SA 4.0. Our repository therefore ships no images or point clouds, only sample tokens, metrics and small renders of our own outputs. The nuScenes frames in the figures of this report are used under that licence, with attribution to the nuScenes authors. Scene statistics used for scene selection are in Table 1b.

LiDAR ground truth (Tracks A and B). For every keyframe and camera the keyframe LiDAR sweep is transformed through LiDAR to ego (at the LiDAR timestamp), ego to world, world to ego (at the camera timestamp) and ego to camera, using the `calibrated_sensor` and `ego_pose` records. Points with $1\,\text{m} < z < 80\,\text{m}$ that fall inside the image are kept; when several points land on one pixel the smallest depth wins (z-buffering). This gives 2,424 sparse depth images with 3020 to 4521 points per image depending on the

camera	pts/img	median z [m]	frac 1-20 m	frac 20-40 m	frac 40-80 m
CAM_FRONT	3020	11.9	0.69	0.23	0.09
CAM_FRONT_LEFT	3344	9.2	0.85	0.13	0.03
CAM_FRONT_RIGHT	3189	10.7	0.70	0.22	0.09
CAM_BACK	4521	9.8	0.74	0.19	0.07
CAM_BACK_LEFT	4000	8.4	0.87	0.12	0.01
CAM_BACK_RIGHT	3614	10.8	0.70	0.23	0.07

(a) LiDAR ground truth per camera: points per image, median depth, share of points per depth bin.

scene	keyframes	annotations	moving (velocity)	frac moving
scene-0916	41	2,424	186	0.08
scene-0655	41	2,373	213	0.09
scene-0757	41	604	294	0.49
scene-1100	40	956	492	0.51
scene-0796	40	716	588	0.82
scene-1077	41	909	701	0.77
scene-0553	41	1,991	796	0.40
scene-0103	40	2,090	968	0.46
scene-0061	39	4,699	1,017	0.22
scene-1094	40	1,776	1,191	0.67

(b) Moving annotations per scene (box velocity above 0.5 m/s), sorted; scene-0916 is the 3DGS scene.

Table 1: Data statistics of nuScenes-mini used by the protocol.

camera, a median depth between 8.4 and 11.9 m and 69 to 87% of the points below 20 m (Table 1a). The projection takes 27 s on the CPU.

Metrics and clipping. Predictions are read at the LiDAR pixels, clipped to $[1, 80]$ m and scored with AbsRel, SqRel, RMSE, $\text{RMSE}_{\log}$ and the threshold accuracies δ_k (fraction with $\max(d/\hat{d}, \hat{d}/d) < 1.25^k$). The metrics are accumulated over all points and not averaged per image. We clip because in a first run without clipping a handful of sky-like predictions, which Depth Pro clamps at 10 km, dominated the squared metrics. The fraction of LiDAR pixels whose unclipped prediction exceeds 80 m is reported next to the metrics. "Raw" is the model's metric output. "Median-scaled" multiplies each image by $\text{median}(d_{\text{LiDAR}})/\text{median}(\hat{d})$ before scoring, so what remains is shape error only. The ratio of the two tells how much of the raw error a single scalar per image removes. Confidence intervals are 95% percentile bootstrap intervals (2000 resamples, fixed seed) over images, sequences or test pairs. Paired intervals are computed over per-item differences, and we report them together with the fraction of items in which the first method is lower.

Track A models. Depth Pro [1] runs in half precision at its native 1536×1536 input. One forward pass gives both the own-focal depth and the known-focal depth, because the canonical inverse depth is scaled by W/f, where f is either the predicted or the true f_x in pixels. We checked the known-focal path against the library's own `infer(f_px)` on cpu fp32 and the maximum absolute difference is 0.0 m. Depth Anything V2 metric outdoor Large [13] runs through transformers with its default preprocessing. MoGe-2 [10] is run with its own focal estimate and with the true focal length. Timing is the median model time per image on the NVIDIA L4, including pre- and post-processing and excluding JPEG decoding.

Track B alignment. VGGT-1B [9] runs in bfloat16 on images resized to 518×294, and its depth is upsampled bilinearly to the LiDAR pixels. We use two settings. The first is the six-camera rig at one timestamp with CAM_FRONT as reference, which has wide baselines and little overlap. The second is five consecutive CAM_FRONT keyframes with the centre frame as reference, which is forward motion. Each setting has 40 sequences, four per scene, spread over the scene. Since VGGT has no metric scale, its reference-view depth is aligned in three ways: with one scale per sequence (the median estimator applied to the LiDAR pixels of all views of the sequence), with a per-image median scale, and with the scale of a similarity transform fitted between its predicted camera centres and the nuScenes camera centres [8]. Depth Pro runs on the same reference image, so the paired comparison uses identical pixels. Relative poses between the reference and each other view are compared with nuScenes ego poses and calibration. Rotation error is computed over all pairs. Translation direction error is computed only over pairs whose true baseline exceeds 0.1 m, because the direction of a zero vector is undefined, and the number of excluded stationary pairs is reported.

Track C scene and split. We count moving annotations per scene with the devkit box velocity (threshold 0.5 m/s) and pick the scene with the fewest, scene-0916 (186 of 2,424 annotations, 8%, Table 1b). All six cameras of every keyframe are used at half resolution (800×450). The held-out set is every 8th timestamp (offset 4) with all six cameras, which gives 30 views, and the remaining 216 views are used for training. Random initialisation draws uniform points in a box around the trajectory and gives them random colours.

Table 2: Track A on all 2,424 images (points in $[1, 80]$ m). Raw is the metric output; median-scaled applies one scalar per image. ms/img is the median model time per 1600×900 image on the NVIDIA L4; "true f" feeds the true focal length.

model	alignment	AbsRel	SqRel	RMSE [m]	$\text{RMSE}_{\log}$	δ_1	δ_2	δ_3	ms/img
Depth Pro	raw	0.257	2.123	7.760	0.488	0.525	0.800	0.894	599
	median-scaled	0.178	1.629	6.725	0.314	0.760	0.902	0.951	
Depth Pro (true f)	raw	0.277	2.448	8.579	0.553	0.460	0.727	0.858	599
	median-scaled	0.178	1.629	6.725	0.314	0.760	0.902	0.951	
DA-V2 outdoor L	raw	0.582	5.133	8.957	0.524	0.199	0.604	0.787	365
	median-scaled	0.194	1.643	7.446	0.336	0.715	0.871	0.937	
MoGe-2	raw	0.193	1.366	6.025	0.316	0.731	0.915	0.955	188
	median-scaled	0.151	1.277	6.085	0.292	0.813	0.922	0.956	
MoGe-2 (true f)	raw	0.168	1.258	5.932	0.308	0.812	0.922	0.953	188
	median-scaled	0.152	1.280	6.064	0.293	0.812	0.922	0.957	

LiDAR initialisation aggregates the LiDAR sweeps of the training timestamps in the world frame, colours the points by projection, and voxel-downsamples and caps them. The third kind of variant adds a depth loss to LiDAR initialisation. This loss is the relative L1 between the rendered expected depth and the cached Depth Pro map of the view, scale-aligned per image to LiDAR as in Track A. It is applied on pixels with accumulated alpha above 0.5 and target depth in $(0.5, 80)$ m, with a weight of 0.05 or 0.2. The photometric loss is $0.8\,L_1 + 0.2\,(1 - \text{SSIM})$, and gsplat's `DefaultStrategy` densifies from step 500 to half the budget. We train for 7000 and for 15000 steps and report per-view PSNR, SSIM and LPIPS (AlexNet) on the held-out views. The 7000-step runs are repeated with 3 seeds.

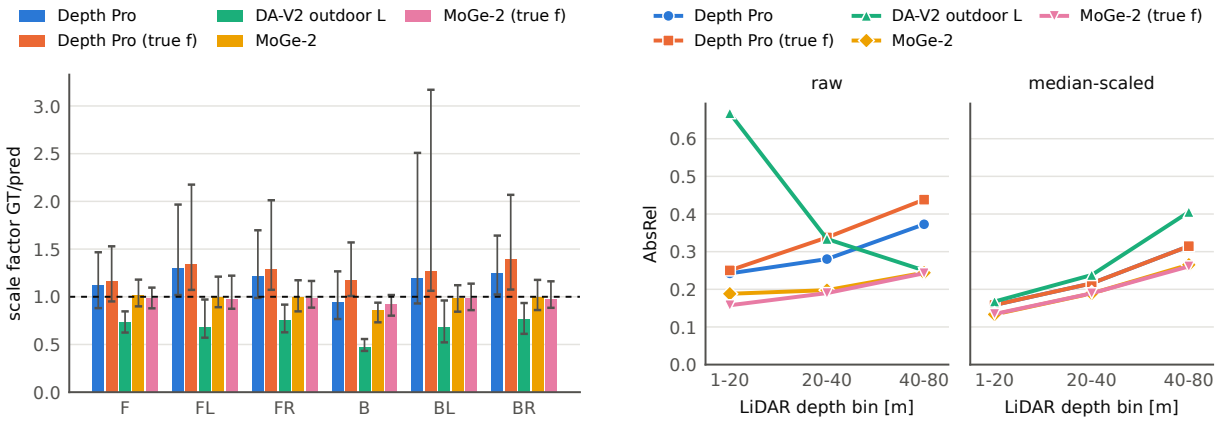
Track D coverage masking. SHARP [7] predicts Gaussians from one CAM_FRONT keyframe per scene. It receives the true focal length and assumes the principal point at the image centre. We render the Gaussians with SHARP's own gsplat-based renderer at three poses: the identity pose, the true relative pose of the next 12 Hz sweep, and the true relative pose of the next keyframe (0.50 s later). The poses come from nuScenes ego poses and calibration. Renders are compared with the real frame at 800×450. PSNR is computed on pixels with alpha above 0.5 and SSIM is averaged over the same pixels. For LPIPS the uncovered pixels are filled from the real frame, which makes LPIPS slightly optimistic when coverage is low, so we report coverage as well.

Track E data and budget. All CAM_FRONT keyframes and sweeps (2,342 frames) are resized to 128×64 and paired with their successor inside the scene (median spacing 0.10 s). Scenes scene-0655, scene-1094 are held out, which gives 1,865 training and 467 test pairs. A 12.4M-parameter UNet (channels 64, 128, 256, attention at the coarsest level) receives the noisy target and the current frame and predicts the flow-matching velocity for the linear path $x_t = (1 - t)x_0 + tx_1$ [6], with 10% condition dropout. The same UNet with zero noise and $t = 0$ is the L2 regression baseline. Both are trained with AdamW (batch 64, learning rate 0.0002, EMA 0.999) under a 40-minute wall-clock cap while sharing the GPU with each other, so only 2,150 of the requested 12,000 steps were done. For evaluation we draw one Euler, 50 steps sample per test pair and report PSNR and LPIPS against the real next frame, next to the copy-current-frame baseline. The distributed code path was smoke-tested with 2 gloo ranks on the single GPU for 20 steps. This is not a multi-GPU result.

4 Results

4.1 Track A: monocular metric depth versus LiDAR

Table 2 gives the overall numbers. For all three models part of the raw metric error is a scale error per image, but the size of that part differs a lot. Median scaling lowers Depth Pro's AbsRel from 0.257 to 0.178 (31% of the raw error) and raises its δ_1 from 0.525 to 0.760. For Depth Anything V2 the drop is from 0.582 to 0.194 (67%), and for MoGe-2 from 0.193 to 0.151 (22%). MoGe-2 is the most accurate of the three both before and after scaling. After scaling, Depth Pro is slightly ahead of Depth Anything V2, although their



(a) Median per-image scale factor GT/pred per camera (whiskers p10 to p90; dashed line: 1.0, already metric).

(b) AbsRel per depth bin, raw and median-scaled. The two Depth Pro variants coincide after scaling by construction.

Figure 1: Track A: where the metric scale is off (a) and how the error changes with distance (b). Cameras: F front, FL/FR front left/right, B back, BL/BR back left/right.

raw errors are far apart. The per-image bootstrap means agree with the point-weighted values (Depth Pro 0.255 [0.250, 0.260] raw and 0.177 [0.173, 0.182] scaled, Depth Anything V2 0.560 [0.549, 0.571] and 0.193 [0.189, 0.197], MoGe-2 0.187 [0.183, 0.190] and 0.151 [0.147, 0.156]). Only a small fraction of LiDAR pixels has an unclipped prediction beyond 80 m: 0.08% for Depth Pro, 0.00% for Depth Anything V2 and 0.10% for MoGe-2. On the NVIDIA L4 the median time per image is 599 ms for Depth Pro, 365 ms for Depth Anything V2 and 188 ms for MoGe-2.

Scale and focal length. Figure 1a and Table 3a show the median scale factor per camera. With its own focal estimate, Depth Pro predicts depths that are too near on the five long-focal cameras (scale 1.12 to 1.30) and close to metric on the back camera (0.94). Its focal estimate is above the true nuScenes focal on every camera, by a factor of 1.04 to 1.28. The largest ratio is on the back camera, whose true focal (809 px) is shorter than that of the other five (about 1266 px). We expected the true focal to help. It does not: AbsRel rises from 0.257 to 0.277 and δ_1 falls from 0.525 to 0.460, while the median-scaled numbers are identical by construction. The scale column explains why. With the true focal the scale factor is similar on all six cameras (1.16 to 1.39), which means that the model's canonical depth is consistently too near on this data and that its over-estimated focal cancels part of that. MoGe-2 behaves the way one would expect. Its focal estimate is between 0.96 and 1.22 times the true focal, its scale factors lie between 0.85 and 1.01, and the true focal lowers its raw AbsRel from 0.193 to 0.168. Depth Anything V2 metric outdoor, fine-tuned on Virtual KITTI, predicts too far on every camera (scale 0.47 to 0.77, lowest on the back camera with 0.47).

Cameras and distance. The front camera is the easiest for all three models, raw and scaled, and after scaling the side cameras are the hardest (Tables 4a and 3b). Depth Pro has a scaled AbsRel of 0.124 on the front group and 0.190 on the side group, Depth Anything V2 has 0.140 and 0.207, and MoGe-2 has 0.112 and 0.163. Before scaling the picture is mixed. Depth Pro is again worst on the side group (0.270 against 0.191 on the front), but Depth Anything V2 and MoGe-2 are worst on the back camera (1.034 and 0.289), where their scale is furthest off. Errors grow with distance (Figure 1b, Table 4b). Depth Pro's scaled AbsRel is 0.158 below 20 m, 0.216 between 20 and 40 m and 0.315 beyond 40 m, with RMSE from 2.99 to 20.43 m. The raw error of Depth Anything V2 is the one exception. It falls with distance (0.668 to 0.250) because a global over-estimate of depth hurts most where the true depth is small. After scaling it grows with distance like the others.

model	quantity	F	FL	FR	B	BL	BR
nuScenes calibration	true focal [px]	1266	1273	1261	809	1257	1260
Depth Pro	median scale GT/pred	1.122	1.298	1.212	0.943	1.195	1.249
	pred/true focal	1.046	1.039	1.092	1.276	1.087	1.123
Depth Pro (true f)	median scale GT/pred	1.165	1.342	1.287	1.169	1.267	1.390
DA-V2 outdoor L	median scale GT/pred	0.728	0.685	0.757	0.474	0.677	0.765
MoGe-2	median scale GT/pred	1.012	0.993	0.995	0.855	0.982	0.989
	pred/true focal	0.958	0.974	0.998	1.225	1.010	0.992
MoGe-2 (true f)	median scale GT/pred	0.984	0.970	0.984	0.923	0.988	0.974

(a) Median per-image scale factor GT/pred and focal estimate over true focal. A scale of 1.0 means the metric output is already on the LiDAR scale.

model	F	FL	FR	B	BL	BR
Depth Pro	0.191 / 0.124	0.286 / 0.182	0.237 / 0.204	0.259 / 0.177	0.301 / 0.174	0.251 / 0.202
Depth Pro (true f)	0.202 / 0.124	0.310 / 0.182	0.275 / 0.204	0.236 / 0.177	0.326 / 0.174	0.309 / 0.202
DA-V2 outdoor L	0.399 / 0.140	0.514 / 0.202	0.432 / 0.215	1.034 / 0.191	0.542 / 0.190	0.412 / 0.223
MoGe-2	0.122 / 0.112	0.169 / 0.161	0.175 / 0.179	0.289 / 0.140	0.195 / 0.152	0.169 / 0.164
MoGe-2 (true f)	0.128 / 0.123	0.175 / 0.163	0.161 / 0.178	0.190 / 0.134	0.189 / 0.150	0.153 / 0.166

(b) AbsRel raw / median-scaled.

Table 3: Track A per camera (F front, FL/FR front left/right, B back, BL/BR back left/right).

model	group	AbsRel raw	scaled	δ_1 raw	scaled	points
Depth Pro	front	0.191	0.124	0.633	0.845	1,219,883
	side	0.270	0.190	0.495	0.739	5,715,488
	back	0.259	0.177	0.549	0.771	1,826,681
Depth Pro (true f)	front	0.202	0.124	0.595	0.845	1,219,883
	side	0.306	0.190	0.411	0.739	5,715,488
	back	0.236	0.177	0.525	0.771	1,826,681
DA-V2 outdoor L	front	0.399	0.140	0.242	0.821	1,219,883
	side	0.477	0.207	0.222	0.697	5,715,488
	back	1.034	0.191	0.097	0.699	1,826,681
MoGe-2	front	0.122	0.112	0.860	0.880	1,219,883
	side	0.178	0.163	0.779	0.790	5,715,488
	back	0.289	0.140	0.493	0.843	1,826,681
MoGe-2 (true f)	front	0.128	0.123	0.889	0.860	1,219,883
	side	0.170	0.164	0.809	0.790	5,715,488
	back	0.190	0.134	0.769	0.847	1,826,681

(a) Per camera group (front: F; side: FL, FR, BL, BR; back: B).

model	bin	AbsRel raw	scaled	RMSE raw	scaled	points
Depth Pro	1-20 m	0.242	0.158	3.52	2.99	6,653,105
	20-40 m	0.280	0.216	10.22	8.81	1,599,744
	40-80 m	0.373	0.315	23.38	20.43	509,203
Depth Pro (true f)	1-20 m	0.250	0.158	3.58	2.99	6,653,105
	20-40 m	0.338	0.216	11.55	8.81	1,599,744
	40-80 m	0.438	0.315	26.08	20.43	509,203
DA-V2 outdoor L	1-20 m	0.668	0.168	6.96	2.50	6,653,105
	20-40 m	0.334	0.238	12.10	9.21	1,599,744
	40-80 m	0.250	0.405	16.98	24.62	509,203
MoGe-2	1-20 m	0.188	0.133	2.82	2.49	6,653,105
	20-40 m	0.198	0.189	8.22	8.28	1,599,744
	40-80 m	0.244	0.265	17.56	18.46	509,203
MoGe-2 (true f)	1-20 m	0.158	0.134	2.62	2.51	6,653,105
	20-40 m	0.190	0.189	8.11	8.29	1,599,744
	40-80 m	0.243	0.261	17.57	18.29	509,203

(b) Per LiDAR depth bin (RMSE in m).

Table 4: Track A broken down by camera group and by depth bin, raw and median-scaled, with the number of LiDAR points.

4.2 Track B: feed-forward multi-view reconstruction versus the monocular prior

VGGT is not metric. Its raw depth is off by a large constant factor (median sequence scale 23.4 for the rig and 29.7 for the temporal window, raw AbsRel 0.901), so we only compare after alignment (Table 5). With one scale per sequence VGGT reaches an AbsRel of 0.135 on the rig and 0.118 on the temporal window. Depth Pro with its per-image median scale reaches 0.135 and 0.128 on the same pixels. The paired difference, VGGT minus Depth Pro so that negative values favour VGGT, is -0.001 [-0.024, 0.017] for the rig and -0.010 [-0.026, 0.007] for the temporal window, and both intervals include zero. If VGGT also gets a per-image scale, the differences are -0.012 [-0.029, 0.003] and -0.005 [-0.020, 0.012]. VGGT is ahead on average and in 70% of the temporal sequences, while the rig is a tie (VGGT lower in 42%). On this data, then, multi-view context does not beat a good monocular prior by a margin that a sample of this size can resolve. Scaling VGGT with the similarity transform fitted to its camera centres works much worse (0.607 and 0.321). The baselines are short (mean 0.81 m inside the rig, 4.24 m between keyframes), so the fitted scale is noisy.

Poses. Table 6 gives the relative pose errors. On the temporal window VGGT's relative rotations are accurate (median 0.14 degrees, mean 0.31) and the translation directions are good too (median 2.0 degrees over the 120 moving pairs, with 40 stationary pairs excluded). On the six-camera rig the rotations are off by several degrees (median 4.65, mean 8.12) and the translation directions are poor (median 35.9 degrees). The nuScenes cameras overlap very little, so this is not surprising.

Table 5: Track B: AbsRel, δ_1 and RMSE on the LiDAR pixels of the reference view, mean over 40 sequences per setting with 95% bootstrap intervals. VGGT and Depth Pro are scored on identical pixels.

alignment / model	AbsRel [95% CI]	δ_1	RMSE [m]
(i) 6-camera rig, one timestamp, reference CAM_FRONT			
VGGT raw (relative scale)	0.901 [0.898, 0.903]		
VGGT, one scale per sequence	0.135 [0.120, 0.149]	0.818 [0.787, 0.848]	6.80 [6.08, 7.57]
VGGT, sim(3) scale from cameras	0.607 [0.517, 0.699]		
VGGT, per-image median scale	0.124 [0.108, 0.140]	0.854 [0.828, 0.880]	6.35 [5.64, 7.10]
Depth Pro raw (metric)	0.188 [0.166, 0.211]		
Depth Pro, per-image median scale	0.135 [0.116, 0.161]	0.821 [0.767, 0.863]	6.74 [6.12, 7.36]
Depth Pro raw, true focal	0.200 [0.178, 0.223]		
(ii) 5 consecutive CAM_FRONT keyframes, reference = centre frame			
VGGT raw (relative scale)	0.901 [0.898, 0.904]		
VGGT, one scale per sequence	0.118 [0.097, 0.146]	0.867 [0.818, 0.903]	5.88 [5.18, 6.68]
VGGT, sim(3) scale from cameras	0.321 [0.208, 0.452]		
VGGT, per-image median scale	0.123 [0.100, 0.153]	0.862 [0.808, 0.902]	5.93 [5.19, 6.74]
Depth Pro raw (metric)	0.186 [0.162, 0.211]		
Depth Pro, per-image median scale	0.128 [0.112, 0.151]	0.831 [0.778, 0.872]	6.45 [5.88, 7.07]
Depth Pro raw, true focal	0.191 [0.170, 0.214]		

setting	pairs	rot mean	rot median	trans mean	trans median	excluded	baseline [m]	seq. scale
(i) rig	200	8.12	4.65	48.6	35.9	0	0.81	23.4
(ii) temporal	160	0.31	0.14	2.3	2.0	40	4.24	29.7

Table 6: Relative pose of VGGT against nuScenes ego poses and calibration: rotation error and translation direction error in degrees. Translation direction is evaluated only where the true baseline exceeds 0.1 m; "excluded" counts the stationary pairs; "seq. scale" is the median sequence scale factor.

variant	steps	PSNR	PSNR, seeds	LPIPS, seeds	SSIM	LPIPS	init	final	train min	ms/view	PSNR train
random init	7k	13.40 ± 3.78	13.78 ± 0.42 (n=3)	0.744 ± 0.012	0.521	0.758	200,000	206,811	1.5	1.9	20.01
LiDAR init	7k	16.29 ± 2.75	15.91 ± 0.37 (n=3)	0.590 ± 0.009	0.587	0.580	300,000	576,114	2.9	3.5	21.24
LiDAR init + depth (w=0.05)	7k	17.54 ± 2.66	17.46 ± 0.10 (n=3)	0.522 ± 0.003	0.599	0.525	300,000	674,323	3.3	3.9	21.66
LiDAR init + depth (w=0.2)	7k	18.51 ± 2.77	18.43 ± 0.07 (n=3)	0.498 ± 0.001	0.608	0.497	300,000	736,227	3.5	4.2	21.48
random init	15k	12.36 ± 2.45	not measured	not measured	0.512	0.729	200,000	427,153	4.4	2.6	20.61
LiDAR init	15k	14.04 ± 3.42	not measured	not measured	0.522	0.605	300,000	948,796	8.4	5.0	22.44
LiDAR init + depth (w=0.05)	15k	16.94 ± 2.79	not measured	not measured	0.574	0.469	300,000	1,337,190	11.6	7.2	23.43
LiDAR init + depth (w=0.2)	15k	18.32 ± 2.63	not measured	not measured	0.584	0.417	300,000	1,516,755	12.8	8.3	23.32

(a) Held-out metrics, number of Gaussians (initial and final), training minutes and render time per view on the NVIDIA L4. PSNR is mean ± standard deviation over views for seed 0; "PSNR train" is on a subset of training views. The seed columns give mean ± sample standard deviation of the per-run means over the seeds that were run (n in brackets).

Table 7: Track C ablation on scene-0916, 30 held-out views at 800×450, NVIDIA L4.

4.3 Track C: 3D Gaussian Splatting on one driving scene

Table 7a and Figure 2 show the ablation. The order of the variants is the same at both step budgets. Random initialisation is worst (13.40 dB at 7k). Aggregated LiDAR initialisation is better on all three metrics (16.29 dB, a gain of 2.89 dB). Depth supervision from the scale-aligned Depth Pro maps improves held-out PSNR, SSIM and LPIPS again, and the larger weight does better: 17.54 dB at weight 0.05 (1.26 dB and -0.054 LPIPS over LiDAR initialisation) and 18.51 dB at weight 0.2 (2.22 dB and -0.083 LPIPS). The 15k runs keep this order (12.36, 14.04, 16.94 and 18.32 dB), and the best variant improves its LPIPS to 0.417. Held-out PSNR, however, is lower at 15k than at 7k for every variant, while training-view PSNR goes up (21.48 to 23.32 dB for the best variant). The longer runs also end with about twice as many Gaussians (736,227 to 1,516,755) and take 12.8 instead of 3.5 minutes on the NVIDIA L4.

We repeated the 7k runs with 3 seeds. The seed means are 13.78, 15.91, 17.46 and 18.43 dB, with standard deviations between 0.07 and 0.42 dB. That spread is small against the gaps between the variants, and the order of the four variants is the same in every seed. The 15k runs were done with seed 0 only.

The absolute quality is low for 3DGS, and the numbers should be read with that in mind. The held-out views are 0.5 s, which is several metres, away from the nearest training views. Every camera has its own exposure, moving people and cars are modelled as static and leave ghosts, and the training views

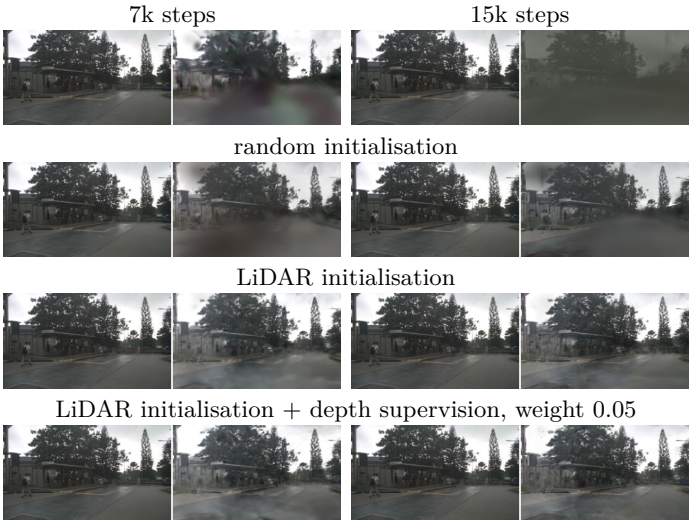


Figure 2: Track C, held-out CAM_FRONT view of the first held-out timestamp, seed 0. Each panel shows the real frame (left) and the render (right). nuScenes images, CC BY-NC-SA 4.0.

target pose	PSNR [95% CI]	SSIM [95% CI]	LPIPS [95% CI]	coverage	motion [m]
identity (input pose)	41.62 [40.69, 42.52]	0.991 [0.987, 0.995]	0.017 [0.014, 0.020]	1.00	0.00
next 12 Hz sweep	26.80 [24.70, 28.87]	0.773 [0.683, 0.858]	0.096 [0.071, 0.122]	1.00	0.48
next keyframe	20.42 [18.71, 22.09]	0.592 [0.501, 0.684]	0.231 [0.179, 0.290]	1.00	2.82

(a) Track D: SHARP Gaussians from one CAM_FRONT keyframe per scene (10 images), rendered at three poses and compared with the real frame on covered pixels. Motion is the mean camera translation.

method	PSNR [95% CI]	LPIPS [95% CI]	PSNR mean ± std (seeds)	LPIPS mean ± std (seeds)
flow matching, one Euler sample	23.82 [23.61, 24.04]	0.099 [0.096, 0.102]	23.80 ± 0.10 (n=3)	0.098 ± 0.006
regression UNet (L2)	24.74 [24.52, 24.98]	0.121 [0.119, 0.123]	24.92 ± 0.19 (n=3)	0.120 ± 0.003
copy current frame	24.38 [24.09, 24.69]	0.040 [0.038, 0.042]	24.38 ± 0.00 (n=3)	0.040 ± 0.000

(b) Track E on 467 held-out pairs from scenes scene-0655, scene-1094: PSNR and LPIPS against the real next frame. The seed columns give mean ± sample standard deviation of the per-run means over seeds.

Table 8: Synthesis tracks, means with 95% bootstrap intervals.

cover the scene only once. Training-view PSNR is far below what dense captures reach, so the model is under-constrained rather than over-fitted.

4.4 Track D: single-image Gaussians rendered at the true next pose

SHARP's Gaussians re-render the input view almost perfectly (PSNR 41.62 dB, LPIPS 0.017 at the identity pose, Table 8a), so the pipeline itself is sound. Quality then drops quickly as the true camera motion grows. At the next 12 Hz sweep (0.48 m of forward motion on average) we measure 26.80 dB and LPIPS 0.096, and at the next keyframe (2.82 m) 20.42 dB and LPIPS 0.231. Coverage is 1.00 in both cases, because the ego vehicle drives into the part of the scene that the Gaussians already cover. Prediction takes 5496 ms per image and a render 22.0 ms per view on the NVIDIA L4. We did not measure where in the image the error sits. Likely sources are content that the input photo could not see, such as the areas behind cars and poles and the road under the ego vehicle, and moving objects, which a single-image model freezes at their position at time t. With ten images, one per scene, the sample is too small to say more than this ordering. Figure 3a shows two next-keyframe renders.



(a) Track D: real next keyframe (left) and SHARP Gaussians of the previous keyframe rendered at its true pose (right), scenes 0061 and 0655.



(b) Track E: current frame (top row), real next frame (middle) and one flow-matching sample (bottom), four held-out pairs at 128×64.

Figure 3: Synthesis tracks. nuScenes images, CC BY-NC-SA 4.0.

Table 9: Wall-clock time of every job on the NVIDIA L4 (LiDAR projection on the CPU).

job	size	minutes (NVIDIA L4)
LiDAR GT projection (CPU)	2,424 images	0.4
Track A DA-V2 outdoor L	2,424 images, 365 ms/img median	14.9
Track A Depth Pro	2,424 images, 599 ms/img median	26.5
Track A MoGe-2	2,424 images, 188 ms/img median	8.0
Track C 3DGS, 4 variants (seed 0)	7,000 steps each, 33.3 to 77.7 steps/s	11.5 (sum)
Track C 3DGS, 4 variants (seed 0)	15,000 steps each, 19.5 to 56.2 steps/s	37.5 (sum)
Track C 3DGS, additional seeds	8 runs	22.4 (sum)
Track B VGGT (+ Depth Pro on reference views)	80 sequences	2.0
Track E train fm	2,150 steps, 0.9 steps/s (GPU shared with the other training)	40.0
Track E train regression	2,150 steps, 0.9 steps/s (GPU shared with the other training)	40.0
Track E training, additional seeds	4 runs	160.1 (sum)
Track E eval (sampling only)	467 pairs, 173.8 ms/sample	1.4
Track E DDP smoke (torchrun, gloo, 2 ranks on one GPU)	20 steps, world size 2	0.08
Track D SHARP	10 images, 5496 ms predict, 22.0 ms render per view	n/a

4.5 Track E: next-frame synthesis with conditional flow matching

Table 8b and Figure 3b give the result. Under the fixed 40-minute budget on a shared GPU, one Euler sample from the flow-matching UNet is perceptually closer to the real next frame than the output of the regression UNet (LPIPS 0.099 versus 0.121, paired difference -0.022 [-0.025, -0.019], interval excluding zero). Its PSNR is lower (23.82 versus 24.74 dB), which is the usual trade-off between a sharp sample and a blurry mean. The copy baseline reaches 24.38 dB and LPIPS 0.040. The regression UNet is slightly above it in mean PSNR but far behind in LPIPS, and the flow-matching sample is behind it on both metrics (paired differences of -0.56 [-0.73, -0.40] dB in PSNR and 0.059 [0.057, 0.061] in LPIPS). Consecutive frames are only 0.10 s apart, so copying is a strong baseline, and a 12.4M-parameter model trained for 2,150 steps does not learn enough motion to beat it perceptually. The FID of unconditional samples against the held-out frames is 390.3 (467 generated frames at 128×64, upsampled for the Inception network). It is poor, and we report it only to show that the model was not simply memorising the condition. Sampling costs 173.8 ms per frame on the NVIDIA L4. Over 3 seeds the flow-matching samples reach 23.80 ± 0.10 dB and LPIPS 0.098 ± 0.006, and the regression model 24.92 ± 0.19 dB and 0.120 ± 0.003, so the picture does not depend on the seed.

4.6 Runtimes

Table 9 lists every job with its wall-clock time on the NVIDIA L4. The Track A, B and C jobs ran one after another on an otherwise idle GPU, with two exceptions. A few short smoke tests shared the GPU during the Depth Pro rerun, which is why we report the median time per image, and the two Track E trainings of each seed shared the GPU with each other.

5 Discussion

Scale versus shape. The raw versus median-scaled split shows how much of the error one scalar per image removes on nuScenes: 31% of Depth Pro's AbsRel, 67% of Depth Anything V2's and 22% of MoGe-2's. Before scaling the three models are far apart (0.193, 0.257 and 0.582). After scaling they are close (0.151, 0.178 and 0.194). The models therefore differ much more in how well they guess the metric scale than in the shape they recover, and a ranking by raw error is to a large part a ranking of the scale guess. MoGe-2 leads on both counts here. A practical consequence is that any source of one scalar per image, for example a few LiDAR returns, a known object size or a multi-view constraint, is worth more for Depth Pro and Depth Anything V2 than switching between the two. Track B points the same way. A multi-view model without metric scale, given one scalar per sequence, reaches the same AbsRel as the scaled monocular model on identical pixels.

Depth Pro and the true focal length. Depth Pro's depth is proportional to the focal length it uses, $d = f\,/\,(W\,c)$, where c is its canonical inverse depth. Its focal estimate is above the true nuScenes focal on every camera, most of all on the back camera, which has the shortest true focal. Even so, the output with the model's own focal is closer to metric than the output with the true focal. The scale factors explain this. With the true focal all six cameras show a similar under-estimation (scale 1.16 to 1.39), so the canonical depth itself is too near on this data, and the over-estimated focal happens to compensate part of it. To rule out a bug on our side we cross-checked the result against the library's own known-focal path (maximum difference 0.0 m). For this model on this data, the correct intrinsics give a worse raw result overall than the model's own guess. The back camera is the exception (raw AbsRel 0.236 with the true focal against 0.259 with the estimated one), because there the estimated focal over-compensates. We cannot tell from this data which of the two effects, the canonical depth or the focal estimate, reflects the training distribution. MoGe-2 does not show the effect. The true focal lowers its raw AbsRel from 0.193 to 0.168.

Side cameras. After scaling, the side cameras are the hardest for all three models, and for Depth Pro they are the hardest before scaling as well. We think the reason is content rather than optics. Side views are dominated by close, oblique facades, parked cars and vegetation, with many LiDAR returns under 20 m (Table 1a). This is only a hypothesis. Our data cannot separate it from a mismatch in focal length or field of view to the training distribution. One observation speaks against a pure focal effect: after scaling, the back camera, which has the shortest focal, is not the hardest for any of the three models (0.177, 0.191 and 0.140).

Why the geometry prior helps 3DGS. The capture sees the scene once, with views 0.5 s apart. In that setting the photometric loss alone does not constrain the geometry enough, as the random-initialisation result shows. LiDAR points fix where surfaces are, and the aligned monocular map adds a dense target where LiDAR is sparse. Within the two weights we tested, more depth weight gives more gain, at both step budgets. The limit of the prior shows up as well. Training longer raises training-view PSNR but not held-out PSNR, which suggests that the model is limited by what the views constrain (exposure, moving objects and sky are not modelled) and not by the length of the optimisation.

The copy-frame baseline. At 0.10 s spacing and 128×64 resolution the next frame differs from the current one by a small motion. A copy of the current frame is therefore already close to the target in both PSNR and LPIPS, and a generative model has to reproduce the current frame exactly and also move it correctly just to break even. Track D shows a similar pattern. SHARP's geometry handles a 0.48 m step well and a 2.82 m step poorly, because the parts of the next view that the current image does not contain have to be invented. In both tracks the quality of the synthesis ends where the geometry available from the input ends.

6 Limitations

- **Mini split only.** nuScenes-mini has 10 scenes, 404 keyframes and one city pair. Nothing here is a claim about full nuScenes or other data.
- **Seeds.** The 7k 3DGS runs use 3 seeds and the next-frame models 3. The 15k 3DGS runs and the evaluations of the pretrained models were each done once. Bootstrap intervals over views, pairs, images or sequences do not capture seed variance.
- **Static-scene assumption.** 3DGS and SHARP treat the scene as static, so moving people and cars leave ghosts in every variant. We picked the most static scene for 3DGS, but it still contains 186 moving annotations.
- **Reduced resolution.** 3DGS trains and evaluates at 800×450, SHARP is scored at 800×450 and Track E uses 128×64 frames. VGGT sees 518×294 and its depth is upsampled to the LiDAR pixels.
- **Sparse LiDAR ground truth.** There are only 3020 to 4521 points per image, and they are biased towards the near range. Occlusion at depth discontinuities, moving objects and LiDAR-camera time offsets add noise that no model can remove, and z-buffering cannot remove points that are occluded for the camera but visible for the LiDAR.
- **Models as published.** The zero-shot models use their public weights and default preprocessing. Depth Anything V2 metric outdoor is fine-tuned on Virtual KITTI, so its scale reflects KITTI's camera. Depth Pro and SHARP assume the principal point at the image centre, while the calibrated nuScenes principal points are on average 22 px (horizontal) and 30 px (vertical) away from the centre of the 1600x900 images, at most 58 px.
- **Fixed budgets, single machine.** 3DGS uses fixed step budgets without tuning. The Track E models are tiny, trained under a wall-clock cap on a shared GPU and sampled once per pair with plain Euler, and an FID over a few hundred small frames is only indicative. Every timing is from one NVIDIA L4, and the distributed code path was only smoke-tested on it.

7 Conclusion

On this driving data, zero-shot monocular depth models recover shape much better than metric scale. A feed-forward multi-view model with one scale per sequence reaches the same accuracy on identical pixels and also gives accurate relative rotations between consecutive frames. The scale-aligned monocular depth helps Gaussian Splatting from sparse views. Synthesis from a single image or frame holds up only for small changes. A pose change of 0.48 m costs little and one of 2.82 m costs a lot, and at 0.10 s spacing a copy of the current frame is perceptually closer to the next frame than our small generative model. All of this was measured on ten scenes and one GPU with few seeds, so it should be taken as a small, checkable data point and not as a benchmark. Everything can be regenerated with the released scripts.

Reproducibility statement

The code, the measured json files and the scripts that write every table and macro of this report are at `https://github.com/pavanyadava007/depth3d-lab`. There is one script per experiment in `scripts/`, and the json files they write are in `results/`. `paper/make_tables.py` and `paper/make_figs.py` regenerate `paper/tables/`, `paper/numbers.tex` and `paper/figs/`, and Tectonic compiles `paper/main.tex`. The environment is Python 3.10, torch 2.7.1 with CUDA 12.6 wheels and gsplat 1.5.3 on one NVIDIA L4 (23 GB). No number in the prose is typed by hand. Each is a macro written by `make_tables.py`, and a missing result file produces the words "not measured" instead of a value.

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